

Reads over Atom Spaces: A Mathematical View of Context, Retrieval, and Parameter Routing in Language Models

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Abstract

Attention, external retrieval, and parameter-expert routing all compute query-conditioned weighted reads. We provide a common analytical lens based on valued atom spaces $(\Omega, \mathcal{F}, \nu, \nu)$ and the read operator $R(x; \Omega) = \int_{\Omega} \nu(\omega) d\mu_x(\omega)$, which reduces to a softmax-weighted value combination under a counting measure. First, we derive a Lipschitz-margin sufficient condition for an inserted record to enter top- k retrieval. On 384-dimensional semantic embeddings, the condition is vacuous even when insertion succeeds. Second, we derive exact softmax-tail certificates and an error-budgeted truncation rule. On synthetic corpora with $N = 25,000$, the certificate selects 52–86% of atoms, confirming that it certifies approximation error but is not a sublinear retrieval method. Third, signed reference measures permit exact cancellation of a matched external record by an anti-atom; a 3,000-trial stress test shows that this cancellation is highly sensitive to key perturbations.

On CounterFact-500, held-out ROC calibration of a frozen-MiniLM activation threshold ($\theta^* = 0.55$; test AUC 0.829) increases paraphrase activation from 38.2% to 53.2%, while neighborhood false activation rises from 9.6% to 16.1%. Matched-scale GPT-2 experiments place this gated retrieval beside a rank-one parametric reference and GRACE, revealing a generalization–locality trade-off associated with their key spaces. Controlled routing and first-token steering experiments are reported as diagnostics, with their limitations stated explicitly. We do not propose a new editing method or claim superiority; rather, we offer a shared vocabulary and empirical calibrations of what retrieval, metric, and tail arguments can certify.

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Keywords: Language models, Atom spaces, Measure theory, External memory, Vector retrieval, Expert routing, Model editing

1. Introduction

Autoregressive language models encode linguistic abstractions, syntax, and factual knowledge within dense parameter matrices [1, 2, 3]. While gradient-based optimization successfully acquires structural linguistic generalizations, compiling dynamic factual statements directly into static weights presents persistent operational challenges. Real-world facts obsolesce rapidly, require targeted localized updates, and are subject to regulatory erasure mandates (e.g., the “Right to be Forgotten”). Consequently, post-hoc weight modification often triggers catastrophic interference, parameter drift, or high computational costs [4, 5].

To address the rigidity of dense parameter storage, multiple architectural strategies have emerged across distinct sub-disciplines. In-context reasoning caches dynamic token representations directly within the key-value sequence memory of the model [1]. In parallel, retrieval-augmented generation (RAG) and nearest-neighbor language models (k NN-LM) decouple memory from parameters by querying external vector databases of textual passages [6, 7, 8]. Meanwhile, modular computation mechanisms, including Mixture-of-Experts (MoE) architectures and parameter-efficient adapter banks, dynamically route tokens across banks of specialized sub-networks [9, 10, 11]. Although these three paradigms all rely fundamentally on key-value similarity matching [3, 12, 13], they are conventionally formulated using disconnected notations, disparate routing heuristics, and distinct operational semantics.

In this work, we adopt a single analytical lens for in-context attention, external retrieval, and parameter-expert output routing: each is an integral of a value map against a query-conditioned probability measure over a *valued atom space*. Under a counting measure this read is exactly a softmax-weighted combination, so the lens is a change of notation rather than a new algorithm, and we do not claim that the three mechanisms are algorithmically interchangeable. Related expectation-based views of attention appear in kernel attention [13] and sparse-transform attention [14]; our contribution is to use one formalism across the three memory locations, so that insertion, truncation, cancellation, and calibration questions can be posed once and tested on each.

Contributions. The paper makes four contributions, two of which are negative results that we regard as informative. First, Section 3 defines the read operator

34 $R(x; \Omega)$, specializes it to context, corpus, and parameter-expert spaces, and de-
 35 rives from the measure view the requirement that spaces of very different car-
 36 dinality be normalized separately and combined by a gate, with unit-norm keys
 37 across spaces. Second, Section 4 states a checkable Lipschitz-margin sufficient
 38 condition for a newly inserted record to enter the top- k set and then shows, on a
 39 200-record benchmark, that it holds for 0% of records while actual top-1 entry is
 40 100%; the gap quantifies how far worst-case metric arguments are from practice
 41 in 384-dimensional embeddings. Third, Section 5 gives two exact tail bounds for
 42 top- k truncation and an error-budgeted stopping rule, and shows on synthetic cor-
 43 pora up to $N = 25,000$ that the rule attains 100% coverage but selects 52–86% of
 44 atoms, so it is a certificate rather than a speed-up. Fourth, Sections 4 and 6.4 treat
 45 signed reads and calibrated retrieval: signed reference measures permit exact can-
 46 cellation of an external record by a matched anti-atom, a perturbation bound and
 47 3,000 stress trials show cancellation degrades to 72% residual at key perturbation
 48 0.05, and on CounterFact-500 a held-out ROC-calibrated activation threshold im-
 49 proves paraphrase activation by 15 points at a 6.5-point cost in neighborhood false
 50 activation. The same 500 records are edited with a rank-one parametric update and
 51 with GRACE [15] under one scorer, so the retrieval read is placed beside both a
 52 parametric and a memory-based editor at matched scale. The remaining experi-
 53 ments (tri-space routing and GPT-2 first-token steering) are controlled mechanism
 54 studies whose limitations we state where they appear.

55 2. From a Discrete Function to an Integral

56 2.1. The Token-Level Function

57 A language model is conventionally defined as a conditional probability dis-
 58 tribution over a finite vocabulary V of size d , given an input sequence. Fixing a
 59 maximum context capacity c and introducing a null padding symbol $\emptyset$ yields an
 60 augmented vocabulary $\bar{V} = V \cup \{\emptyset\}$. An input sequence is represented as a tuple

$$x = (x_1, \dots, x_c) \in \bar{V}^c,$$

61 where unassigned sequence positions are padded with $\emptyset$. A language model is a
 62 mapping

$$p : \bar{V}^c \rightarrow \Delta^{d-1}, \quad \Delta^{d-1} = \left\{ z \in \mathbb{R}^d : z_i \geq 0, \sum_{i=1}^d z_i = 1 \right\}.$$

63 2.2. Attention as an Integral

64 Let $\phi : \bar{V} \rightarrow \mathbb{R}^e$ denote an embedding map with $\phi(\emptyset) = 0$. Extending the
65 sequence to a step function over the continuous interval $[0, c]$ gives:

$$f_x(t) = \phi(x_{\lceil t \rceil}), \quad t \in (0, c],$$

66 with $f_x(0) = \phi(x_1)$. Let $\alpha_x : [0, c] \rightarrow \mathbb{R}_{\geq 0}$ be an attention weight density satisfy-
67 ing $\int_0^c \alpha_x(t) dt = 1$, with $\alpha_x(t) = 0$ whenever $x_{\lceil t \rceil} = \emptyset$. The context representation
68 vector is:

$$z(x) = \int_0^c \alpha_x(t) f_x(t) dt \in \mathbb{R}^e,$$

69 and the corresponding token prediction distribution is $p(x) = \text{softmax}(Wz(x) +$
70 $b)$. Because f_x is piecewise constant over unit intervals, the continuous integral
71 evaluates algebraically to the standard discrete attention weighted sum [1, 14]:

$$\int_0^c \alpha_x(t) f_x(t) dt = \sum_{i=1}^c \left(\int_{i-1}^i \alpha_x(t) dt \right) \phi(x_i) = \sum_{i=1}^c \alpha_i \phi(x_i).$$

72 This continuous formulation naturally generalizes by replacing the sequence index
73 interval $[0, c]$ with an arbitrary measurable *atom space*.

74 3. Atom Spaces: A Common Object for Context, Retrieval, and Parameters

75 **Definition 1** (Valued Atom Space). A valued atom space is a tuple $(\Omega, \mathcal{F}, \nu, v)$,
76 where $(\Omega, \mathcal{F})$ is a measurable space of elements designated as atoms, ν is a σ -finite
77 reference measure, and $v : \Omega \rightarrow \mathbb{R}^e$ is a measurable value map into a common
78 output representation space.

79 **Definition 2** (Query-Conditioned Kernel and Read Operator). Given a query $x \in$
80 $\mathcal{X}$, let $K_x : \Omega \rightarrow \mathbb{R}_{\geq 0}$ be measurable with

$$0 < Z_x := \int_{\Omega} K_x(\omega) d\nu(\omega) < \infty.$$

81 Then K_x induces a query-conditioned probability measure μ_x on $(\Omega, \mathcal{F})$ via the
82 Radon–Nikodym derivative:

$$d\mu_x(\omega) = \frac{K_x(\omega)}{Z_x} d\nu(\omega).$$

Table 1: Three structural realizations of the unified read operator $R(x; \Omega)$.

Instance	Atom Space Ω	Atom ω	Value Map $v(\omega)$	Operational Equivalent
Context	$\{1, \dots, c\}$	Token index	Embedding $\phi(x_\omega)$	In-context self-attention [1]
Corpus	$\mathcal{D}$ (mutable)	Factual record	Encoded fact vector	Dense retrieval / k NN-LM [7]
Parameters	Θ (expert bank)	Modular expert	Expert output $g_\omega(x) \in \mathbb{R}^e$	MoE-style output routing [9, 10]

83 If $\int_{\Omega} \|v(\omega)\|_2 d\mu_x(\omega) < \infty$, the read conditioned on x is the Bochner integral:

$$R(x; \Omega) = \int_{\Omega} v(\omega) d\mu_x(\omega) \in \mathbb{R}^e.$$

84 When ν is a discrete counting measure over a finite set $\Omega = \{\omega_1, \dots, \omega_N\}$, the
85 Bochner integral evaluates directly to a softmax-weighted sum:

$$R(x; \Omega) = \sum_{i=1}^N \frac{K_x(\omega_i)}{\sum_{j=1}^N K_x(\omega_j)} v(\omega_i).$$

86 As summarized in Table 1, in-context attention, dense retrieval, and expert-output
87 routing are structural realizations of this read operator, differing in their domain
88 of integration and the query-dependence of their values.

89 **Remark 1** (What the measure view does and does not add). *Every result in this*
90 *paper is proved on finite sums and could be stated without measure theory. The*
91 *measure view is retained for three reasons. First, the reference measure ν is the*
92 *natural carrier of per-record mass, so insertion, reweighting, and (via signed ν)*
93 *cancellation become operations on ν alone with K_x and v fixed (Section 3.3). Sec-*
94 *ond, the normalization $Z_x = \int K_x d\nu$ makes explicit that spaces with $|\Omega|$ differing*
95 *by many orders of magnitude cannot share one normalizer (Section 3.2). Third,*
96 *the same object μ_x is what an activation threshold, a top-k truncation, and a rout-*
97 *ing gate all act on, which lets Sections 4–5 reuse one set of definitions across all*
98 *three spaces.*

3.1. The Unified Architecture

Context, external corpus memory, and parameter routing combine into an integrated representation:

$$z(x) = \lambda_{\text{ctx}}(x)R(x; \Omega_{\text{ctx}}) + \lambda_{\text{corpus}}(x)R(x; \Omega_{\text{corpus}}) + \lambda_{\text{params}}(x)R(x; \Omega_{\text{params}}),$$

$$p(x) = \text{softmax}(W_0 z(x) + b_0),$$

where $\lambda(x) \in \Delta^2$ is an input-dependent gating distribution across the three spaces, and $W_0 \in \mathbb{R}^{d \times e}$, $b_0 \in \mathbb{R}^d$ represent the linguistic projection head. For parameter experts, the value is query-dependent, $v_x(\omega) = g_\omega(x)$; the read is therefore an output-space mixture, not an average of parameter tensors.

3.2. System, Calibration, and Tying Principles

Translating the tri-space read formulation into a functional architecture requires addressing scale calibration and representation tying. Because the cardinalities of the respective spaces vary across orders of magnitude—ranging from $|\Omega_{\text{ctx}}| \sim 10^3$ tokens in context and $|\Omega_{\text{params}}| \sim 10^1$ modular experts to $|\Omega_{\text{corpus}}| \sim 10^9$ external passages—normalizing a single joint softmax across their disjoint union $\Omega_{\text{ctx}} \sqcup \Omega_{\text{corpus}} \sqcup \Omega_{\text{params}}$ would cause the most populous space to artificially dominate probability mass. To preserve balanced access, separate intra-space measure normalization is coupled with calibrated convex gating $\lambda(x)$. Crucially, key representations across distinct spaces must share identical geometric scales through unit L_2 -normalization, which prevents differing dot-product magnitudes from biasing the router toward unnormalized representations.

Furthermore, for token generation tasks, the external read vector $z_{\text{corpus}}(x)$ must project coherently onto candidate vocabulary tokens. Letting $E \in \mathbb{R}^{d \times e}$ denote the vocabulary embedding matrix, we tie atom values via $v(\omega) = E[y_\omega]/\|E[y_\omega]\|$ and evaluate final logits as Ez/τ . Under this parameter tying, isolating an atom ω^* aligns $z(x)$ directly with the corresponding target row of the vocabulary matrix.

3.3. Signed Atom Spaces

To support zero-retraining cancellation in external memory, we allow the reference measure ν to be *signed*. Any finite signed measure decomposes as $\nu = \nu^+ - \nu^-$ with $\nu^+ \perp \nu^-$ (Hahn–Jordan); in the finite counting case used throughout this paper this is simply the split of per-atom masses into positive records and negative “anti-atoms”, and no measure-theoretic machinery is needed. Defining the total

130 variation $|\nu| = \nu^+ + \nu^-$, we assume $0 < Z_x^\pm = \int_\Omega K_x d|\nu| < \infty$. The normalized
 131 signed kernel-weighted measure is:

$$d\mu_x^\pm(\omega) = \frac{K_x(\omega)}{Z_x^\pm} (d\nu^+(\omega) - d\nu^-(\omega)).$$

132 Its total variation is normalized to unity, while its signed total mass varies in
 133 $[-1, 1]$; unlike μ_x , it is not a probability measure.

134 4. Mathematical Conditions for Post-Hoc Knowledge Editing

135 Let the query-key kernel be a scaled inner product:

$$K_x(\omega) = \exp\left(\frac{\langle \phi_q(x), \phi_k(\omega) \rangle}{\tau}\right),$$

136 where $\phi_q : \mathcal{X} \rightarrow \mathbb{R}^m$ and $\phi_k : \Omega \rightarrow \mathbb{R}^m$ are query and key encoders, and $\tau > 0$
 137 is a temperature parameter. Base training optimizes ϕ_q and ϕ_k on an initial pool
 138 $\Omega_{\text{train}} \subset \Omega$. Post-hoc editing requires that inserting a novel atom $\omega^* \notin \Omega_{\text{train}}$ at test
 139 time reliably captures probability mass $\mu_x(\omega^*)$ for associated queries x^* , without
 140 modifying encoder parameters.

141 4.1. Kernel Locality

142 **Definition 3** (Metric Locality). *The key encoder ϕ_k is L -Lipschitz continuous with*
 143 *respect to a semantic pseudo-metric d_Ω on Ω if*

$$\|\phi_k(\omega) - \phi_k(\omega')\|_2 \leq L d_\Omega(\omega, \omega') \quad \forall \omega, \omega' \in \Omega.$$

144 **Lemma 1** (Score Stability on Unseen Atoms). *Let ϕ_k be L -Lipschitz with respect*
 145 *to d_Ω . For any training anchor $\omega_0 \in \Omega_{\text{train}}$ and any newly inserted atom $\omega^* \in \Omega$:*

$$\left| \langle \phi_q(x), \phi_k(\omega^*) \rangle - \langle \phi_q(x), \phi_k(\omega_0) \rangle \right| \leq \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0).$$

146 *Proof.* By the Cauchy–Schwarz inequality and the Lipschitz condition:

$$\begin{aligned} \left| \langle \phi_q(x), \phi_k(\omega^*) - \phi_k(\omega_0) \rangle \right| &\leq \|\phi_q(x)\|_2 \|\phi_k(\omega^*) - \phi_k(\omega_0)\|_2 \\ &\leq \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0). \end{aligned}$$

□

147 4.2. The Retrieval Margin Condition and Its Empirical Vacuity

148 Let $\tau_k(x) = \max_{\omega \in \Omega_{\text{train}}}^{(k)} \langle \phi_q(x), \phi_k(\omega) \rangle$ denote the k -th highest score among ex-
 149 isting training atoms. For the novel atom ω^* to enter the top- k retrieved set under
 150 query x , it must satisfy $\langle \phi_q(x), \phi_k(\omega^*) \rangle > \tau_k(x)$. Combining this with Proposi-
 151 tion 1 yields a checkable sufficient condition:

$$\langle \phi_q(x), \phi_k(\omega_0) \rangle - \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0) > \tau_k(x), \quad (1)$$

152 where ω_0 is a training anchor semantically proximal to ω^* . Equation (1) is a suffi-
 153 cient condition for top- k inclusion; successful prediction additionally depends on
 154 retrieved probability mass, value representation, output margin, and specificity on
 155 controls.

156 *Finite-Dataset Diagnostic.* Equation (1) is the type of condition one would hope
 157 to check before inserting a record. We test whether it is ever satisfied on the 200-
 158 record benchmark of Section 6. We instantiate $d_\Omega(\omega, \omega') = \|\phi_0(\omega) - \phi_0(\omega')\|_2$
 159 using frozen MiniLM backbone embeddings. For the residual adapter $\phi(x) =$
 160 $\text{Normalize}(x + 0.1 \cdot \text{MLP}(x))$, an operator-norm calculation combined with the min-
 161 imum adapted norm observed on the dataset gives $L \leq 2.105$, while the maximum
 162 observed pairwise directional ratio is $\hat{L} = 2.236$. Neither establishes the global
 163 premise of Lemma 1; both are finite-sample diagnostics. Evaluating Equation (1)
 164 across all 40 insertion facts using their nearest training anchors ω_0 : actual top-1 en-
 165 try is 100.0%, whereas the margin is positive for 0.0% of records at $k \in \{1, 5, 10\}$.
 166 With inter-fact distances averaging $d_\Omega \approx 0.78$, the penalty $\hat{L} d_\Omega \approx 1.74$ over-
 167 whelms the observed score margin (left-hand side ≈ -1.34 versus $\tau_k \approx 0.35$).

168 We report this as a negative result. Metric-locality arguments of the form
 169 of Lemma 1 appear natural for reasoning about post-hoc insertion, but Cauchy-
 170 Schwarz permits adversarial alignment in 384 dimensions, and on a sparse seman-
 171 tic corpus the resulting worst-case condition certifies nothing even when insertion
 172 succeeds on every record. Any operational guarantee for insertion must therefore
 173 rest on local or directional bounds established independently of the global Lips-
 174 chitz premise; we leave that construction to future work.

175 4.3. Anti-Atoms and Exact External-Read Cancellation

176 The following observation is algebraically immediate; we record it because it
 177 is the exact identity that the stress test of Section 6.8 perturbs.

178 **Proposition 1** (Matched-Pair Cancellation). *Let an atom $\omega^* \in \Omega_{\text{corpus}}$ encode*
 179 *a target fact with tied token value $v^* = E[y^*]/\|E[y^*]\|$. If an anti-atom ω^- is*

180 inserted into the signed atom space with identical key $\phi_k(\omega^-) = \phi_k(\omega^*)$, identical
 181 value $v^- = v^*$, and equal positive masses $m^* := v^+(\{\omega^*\}) = m^- := v^-(\{\omega^-\}) >$
 182 0, then for any query x :

$$R^\pm(x; \{\omega^*, \omega^-\}) = \mathbf{0}.$$

183 *Proof.* Under the signed measure integral of Section 3.3:

$$\int_{\{\omega^*, \omega^-\}} v(\omega) d\mu_x^\pm(\omega) = \frac{K_x(\omega^*)m^*v^* - K_x(\omega^-)m^-v^*}{Z_x^\pm} = \frac{K_x(\omega^*)m^*(v^* - v^*)}{Z_x^\pm} = \mathbf{0}.$$

184 This establishes exact cancellation of the matched pair’s contribution to the signed
 185 external read. $\square$

186 Proposition 1 does not imply that the complete model representation is zero:
 187 background atoms, context, parameter experts, residual model states, and output
 188 bias remain. The positive record and anti-atom remain stored, and the proposition
 189 does not claim removal of knowledge compiled into dense pretrained parameters.

190 **Proposition 2** (Approximate Pair Cancellation). *Let $a = K_x(\omega^*)v^+(\{\omega^*\})$, $b =$*
 191 *$K_x(\omega^-)v^-(\{\omega^-\})$, and let $Z_x^\pm > 0$ be the total-variation normalization. For posi-*
 192 *tive value v and anti-atom value $\tilde{v}$, the residual pair contribution satisfies:*

$$\left\| \frac{av - b\tilde{v}}{Z_x^\pm} \right\|_2 \leq \frac{|a - b| \|v\|_2 + b \|v - \tilde{v}\|_2}{Z_x^\pm}.$$

193 *Proof.* Write $av - b\tilde{v} = (a - b)v + b(v - \tilde{v})$ and apply the triangle inequality, then
 194 divide by $Z_x^\pm$. $\square$

195 Exact cancellation is sensitive to key-induced kernel mismatch, unequal mass,
 196 and value mismatch. Paraphrases affect cancellation only through their effect on
 197 kernel weights; identical keys cancel for all queries.

198 5. Softmax Tail Truncation: An Exact Certificate

199 Evaluating $R(x; \Omega)$ densely over a large corpus is expensive, and practical sys-
 200 tems read only the top- k ranked atoms. We ask what can be *certified* about the
 201 resulting error once the scores are known. We analyze truncation to the exact
 202 ranked top- k elements:

$$\hat{R}_k(x; \Omega) = \sum_{i=1}^k v(\omega_{(i)}) \frac{K_x(\omega_{(i)})}{\sum_{j=1}^k K_x(\omega_{(j)})}.$$

203 In this discrete setting, numerical truncation corresponds to numerical quadra-
 204 ture of the query-conditioned measure against the value map, omitting the low-
 205 probability tail $Z_{\text{tail}} = \sum_{i=k+1}^N e^{s_{(i)}/\tau}$. Dense scoring over N items costs $\mathcal{O}(Ne)$
 206 before the top- k read. When paired with an ANN index, aggregation costs:

$$\mathcal{O}(c \cdot e) + \mathcal{O}(k_{\text{corpus}} \cdot e) + \mathcal{O}(k_{\text{params}} \cdot e).$$

207 Let $N = |\Omega|$, and $V_{\max} := \sup_{\omega \in \Omega} \|v(\omega)\|_2 < \infty$. For query x , let $s_i :=$
 208 $\langle \phi_q(x), \phi_k(\omega_i) \rangle$, with ordering $s_{(1)} \geq \dots \geq s_{(N)}$. Define $Z := \sum_{j=1}^N e^{s_{(j)}/\tau}$ and
 209 $Z_k := \sum_{j=1}^k e^{s_{(j)}/\tau}$. Define the retrieval spectral gap at rank k as $\Delta_k(x) := s_{(1)} -$
 210 $s_{(k+1)} \geq 0$.

211 **Proposition 3** (Softmax Tail Truncation Error Bound). *For any query x and trun-*
 212 *cation depth $k \in \{1, \dots, N - 1\}$, the Euclidean approximation error of top- k*
 213 *truncation satisfies:*

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq 2V_{\max} \cdot (N - k) \cdot \exp\left(-\frac{\Delta_k(x)}{\tau}\right).$$

214 *An alternative bound is:*

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq 2V_{\max} \cdot \left(\frac{N - k}{k}\right) \cdot \exp\left(-\frac{s_{(k)} - s_{(k+1)}}{\tau}\right).$$

215 *Proof.* Decompose the difference between the full read and the top- k truncation:

$$\begin{aligned} R(x; \Omega) - \hat{R}_k(x; \Omega) &= \sum_{i=1}^k v_{(i)} e^{s_{(i)}/\tau} \left(\frac{1}{Z} - \frac{1}{Z_k}\right) + \sum_{i=k+1}^N v_{(i)} \frac{e^{s_{(i)}/\tau}}{Z} \\ &= \frac{Z_k - Z}{ZZ_k} \sum_{i=1}^k v_{(i)} e^{s_{(i)}/\tau} + \frac{1}{Z} \sum_{i=k+1}^N v_{(i)} e^{s_{(i)}/\tau}. \end{aligned}$$

216 Let $Z_{\text{tail}} := Z - Z_k = \sum_{i=k+1}^N e^{s_{(i)}/\tau}$. Notice $\frac{1}{Z_k} \sum_{i=1}^k v_{(i)} e^{s_{(i)}/\tau} = \hat{R}_k(x; \Omega)$, and
 217 define $R_{\text{tail}} := \frac{1}{Z_{\text{tail}}} \sum_{i=k+1}^N v_{(i)} e^{s_{(i)}/\tau}$. Substituting yields:

$$R(x; \Omega) - \hat{R}_k(x; \Omega) = \frac{Z_{\text{tail}}}{Z} (R_{\text{tail}} - \hat{R}_k(x; \Omega)).$$

218 Taking norms and applying the triangle inequality:

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq \frac{Z_{\text{tail}}}{Z} (\|R_{\text{tail}}\|_2 + \|\hat{R}_k(x; \Omega)\|_2) \leq 2V_{\max} \frac{Z_{\text{tail}}}{Z}.$$

219 Since $Z \geq e^{s_{(1)}/\tau}$ and $Z_{\text{tail}} \leq (N - k)e^{s_{(k+1)}/\tau}$:

$$\frac{Z_{\text{tail}}}{Z} \leq \frac{(N - k)e^{s_{(k+1)}/\tau}}{e^{s_{(1)}/\tau}} = (N - k) \exp\left(-\frac{s_{(1)} - s_{(k+1)}}{\tau}\right).$$

220 Substituting proves the first bound. Alternatively, $Z \geq ke^{s_{(k)}/\tau}$ while $Z_{\text{tail}} \leq (N -$
 221 $k)e^{s_{(k+1)}/\tau}$, giving

$$\frac{Z_{\text{tail}}}{Z} \leq \frac{N - k}{k} \exp\left(-\frac{s_{(k)} - s_{(k+1)}}{\tau}\right),$$

222 which proves the second bound. The second bound is tighter whenever $s_{(1)} - s_{(k)} <$
 223 $\tau \log k$. $\square$

224 Both bounds are elementary consequences of $\|R - \hat{R}_k\|_2 \leq 2V_{\max} Z_{\text{tail}}/Z$; their
 225 value is that they are exact (no distributional assumption on scores) and that the
 226 intermediate quantity Z_{tail}/Z is directly computable, which yields the following
 227 stopping rule.

228 **Corollary 1** (Error-Budgeted Exact Truncation). *Let $\rho_k(x) = Z_{\text{tail}}/Z$ be the exact*
 229 *softmax mass outside the first k ranked atoms. For any requested $\epsilon > 0$, define*

$$k_\epsilon(x) = \min\{k \in \{1, \dots, N\} : 2V_{\max}\rho_k(x) \leq \epsilon\}.$$

230 *Then $\|R(x; \Omega) - \hat{R}_{k_\epsilon(x)}(x; \Omega)\|_2 \leq \epsilon$.*

231 *Proof.* By the proof of Proposition 3, $\|R - \hat{R}_k\|_2 \leq 2V_{\max}\rho_k(x)$. Substituting the
 232 condition for $k_\epsilon(x)$ yields the result. $\square$

233 Computing $\rho_k(x)$ requires all N scores, so Corollary 1 is a post-hoc certificate
 234 for a read that has already been scored densely (or whose candidate set is supplied
 235 by an approximate index), not a sublinear retrieval algorithm. Section 6 shows that,
 236 on synthetic corpora, the certified k_ϵ is a large fraction of N ; we regard this as a
 237 useful calibration of how loose exact tail certification is relative to the aggressive
 238 truncation used in practice.

239 6. Empirical Evaluation

240 We report (i) a saturated synthetic sanity check that also feeds the Lipschitz di-
 241 agnostic of Section 4, (ii) empirical behavior of the exact truncation certificate, (iii)
 242 CounterFact-500 retrieval with ROC threshold calibration, (iv) a rank-one para-
 243 metric reference and (v) GRACE at the same scale and under the same scorer,
 244 (vi) tri-space routing, (vii) signed-read cancellation under perturbation, and (viii)
 245 GPT-2 first-token steering. Items (i), (vi), and (viii) are mechanism studies on
 246 templated or small data and are not claims of state-of-the-art editing performance.

247 6.1. Experimental Setup

248 *Corpus and Domain Design.* We construct a benchmark of 200 structured techni-
 249 cal facts distributed across four domains: *Astronomy & Astrophysics*, *Systems &*
 250 *Computing*, *Chemistry & Materials Science*, and *Molecular Biology & Genetics*.
 251 The first 40 items per domain form the adapter-training pool; the remaining 10
 252 items per domain serve as post-hoc insertions. Their labels remain in the global
 253 output vocabulary, making this a transductive label-space evaluation.

254 *Encoder Architecture and Inductive Bias.* We employ a frozen sentence trans-
 255 former (sentence-transformers/all-MiniLM-L6-v2, $d_{\text{in}} = 384$) as the ge-
 256 ometric backbone, paired with a metric-preserving residual adapter:

$$\phi(x) = \text{Normalize}(x + 0.1 \cdot \text{MLP}(x)),$$

257 where the final MLP layer is zero-initialized. Values are tied to token embeddings:
 258 $v(\omega) = E[y_\omega] / \|E[y_\omega]\|$, with predictions evaluated as $p(y | x) \propto \langle z(x), E[y] \rangle / \tau$.

259 *Reproducibility Environment.* Artifacts were generated using Python 3.11.9, Py-
 260 Torch 2.6.0+cu124, NumPy 2.4.6, sentence-transformers 5.7.0, and transformers
 261 4.48.3 on an NVIDIA GeForce RTX 3050 Ti Laptop GPU (4 GB VRAM) with
 262 an AMD Ryzen 7 5800U host. Pinned versions are listed in `requirements.txt`.
 263 The routing sweep reports seeds {40, 41, 42, 43, 44}; the GRACE and rank-one
 264 runs use seed 42.

265 6.2. Synthetic Insertion Sanity Check

266 On the 200-fact templated benchmark, we insert the 40 held-out records into
 267 pools of density {1, 3, 10, 40} items per domain and measure target recall on
 268 canonical queries (efficacy), on paraphrases (generality), and unchanged argmax
 269 on training queries (specificity). Both the trained residual read and a frozen-
 270 MiniLM exact 1-nearest-neighbor baseline score 100.0% on all three metrics at
 271 every density, and inserting all 40 records sequentially without teardown (pool
 272 growing from 161 to 200 atoms) leaves every earlier insertion correctly classified
 273 with zero training-argmax changes across five seeds; target attention mass stays
 274 at $99.68 \pm 0.01\%$ (Table 2). The benchmark is therefore saturated: it establishes
 275 that the read behaves as intended and supplies the embeddings for the Lipschitz
 276 diagnostic of Section 4, but it does not show that the residual adapter improves
 277 over raw embeddings, and we draw no such conclusion. The one non-trivial trend
 278 is that as pool density grows from 1 to 40 the product of adapter spectral norms
 279 rises from 0.469 to 1.003 while target attention mass falls from 99.97% to 99.75%:

Table 2: Synthetic insertion sanity check across 200 templated facts. All classification metrics are saturated for both methods; only the representation dynamics vary with density. Lifelong row: cumulative insertion of all 40 records, five seeds.

Metric	1 / topic	3 / topic	10 / topic	40 / topic
Efficacy / generality / specificity (trained read)	100 / 100 / 100	100 / 100 / 100	100 / 100 / 100	100 / 100 / 100
Efficacy / generality / specificity (frozen 1-NN)	100 / 100 / 100	100 / 100 / 100	100 / 100 / 100	100 / 100 / 100
Target attention mass $\mu_x(\omega^*)$ (%)	99.97	99.97	99.94	99.75
Adapter linear spectral product	0.469	0.818	0.859	1.003
Lifelong (40 sequential inserts, 5 seeds): efficacy / specificity	100.0 $\pm$ 0.0 / 100.0 $\pm$ 0.0; mass 99.68 $\pm$ 0.01%			

Table 3: Softmax tail truncation error $\|R_{\text{exact}} - \hat{R}_k\|_2$ evaluated on context vectors across temperatures τ and cutoffs k . Under unit-normalized embeddings ($V_{\text{max}} = 1$), error is bounded by 2.0 and decays exponentially at sharp operational temperatures ($\tau = 0.05, k = 1 \Rightarrow 3.82 \times 10^{-6}$).

Temperature	Top-1	Top-3	Top-5	Top-10	Top-20
$\tau = 0.05$	3.82×10^{-6}	3.01×10^{-6}	2.62×10^{-6}	2.03×10^{-6}	1.41×10^{-6}
$\tau = 0.10$	0.0182	0.0170	0.0162	0.0145	0.0121
$\tau = 0.50$	0.965	0.700	0.558	0.384	0.245
$\tau = 1.00$	0.988	0.623	0.477	0.327	0.217

the adapter departs further from identity as it is asked to separate more records, at a small cost in mass concentration.

6.3. Convergence of Exact Ranked Top- k Truncation

Table 3 presents empirical truncation error $\|R_{\text{exact}}(x) - \hat{R}_k(x)\|_2$ across temperatures and cutoffs k on the largest atom pool (density 40, 160 active background atoms).

At density 40, $\tau = 0.10$, and $k = 1$, the maximum observed error is 0.0244, and the smaller pointwise bound covers 100% of queries with a minimum slack of 0.0923. Two regimes are visible. At $\tau = 0.05$, mass concentrates on the nearest record and top-1 error is 3.82×10^{-6} . At $\tau \geq 0.50$, mass diffuses across background atoms and premature truncation gives errors near 1.0.

291 *Adaptive error budgets and scaling.* Evaluating Corollary 1 on 64 correlated
 292 synthetic queries with random keys and values in $\mathbb{R}^{128}$, for corpus sizes N
 293 of 1,000, 5,000, 10,000, and 25,000, gives 100% empirical coverage at every
 294 $\epsilon \in \{0.01, 0.05, 0.10\}$, as it must. The informative quantity is the certified depth:
 295 at $N = 25,000$ and $\epsilon = 0.05$, the mean k_ϵ is 13,026 (52.1% of the corpus) at
 296 $\tau = 0.05$ and 21,467 (85.9%) at $\tau = 0.10$. Exact scoring and sorting take 54.6
 297 and 18.2 ms on the benchmark GPU. The certificate is therefore far looser than
 298 the $k \in [5, 50]$ truncations used by practical retrievers; closing that gap requires
 299 distributional assumptions on the score tail that the exact bound does not make.

300 6.4. CounterFact External-Memory Retrieval and Calibrated Activation

301 We evaluate frozen MiniLM retrieval geometry on the first 500 CounterFact
 302 records [4]. Memory keys concatenate the rewrite prompt with its requested target;
 303 evaluation covers 500 rewrite prompts, 1000 paraphrases (2 per record), and 1000
 304 neighborhood prompts (2 per record). A record is *activated* for a query when the
 305 cosine similarity of the top-1 key exceeds a threshold θ (distinct from the softmax
 306 temperature τ).

307 A fixed $\theta = 0.60$ discards most valid paraphrases (38.2% activated). We there-
 308 fore calibrate θ on held-out data: records 500–999 form the calibration split and
 309 records 0–499 the evaluation split. On the calibration split, sweeping θ over posi-
 310 tive queries (rewrites and paraphrases) versus negative distractors (neighborhood
 311 queries) yields ROC AUC 0.817; the same sweep on the evaluation split gives
 312 AUC 0.829. Maximizing Youden’s J on the calibration split selects $\theta^* = 0.55$.

313 Table 4 shows that calibration raises activated paraphrase recall from 38.2%
 314 to 53.2% while raising neighborhood false activation from 9.6% to 16.1%; the
 315 trade-off is roughly 15 points of recall for 6.5 points of false activation, and the
 316 91.0% exact top-1 recall bounds what any threshold can recover. Rewrite recall is
 317 unchanged at 99.2%. FAISS HNSW attains 99.96% exact-top-1 candidate recall
 318 with 12.7 ms construction and 23.5 ms search over all 2,500 queries.

319 6.5. Rank-One Covariance-Regularized Parametric Reference

320 To place the retrieval numbers beside a parametric edit at the same scale, we
 321 implement a rank-one covariance-regularized (R1C) update on GPT-2 (124M) for
 322 the same 500 CounterFact records. The construction follows the ROME update
 323 rule [4]: the edit is applied at the subject span found via tokenizer offsets; a sin-
 324 gle global edit layer (layer 3) is chosen by a causal trace averaged over 50 held-
 325 out records; and the update is $W \leftarrow W + (C^{-1}k)(v^* - k^\top W)^\top / (k^\top C^{-1}k)$ with
 326 $C = \mathbb{E}[kk^\top]$ estimated on 1,500 held-out texts. It departs from the published

Table 4: CounterFact-500 retrieval with uncalibrated ($\theta = 0.60$) and held-out-calibrated ($\theta^* = 0.55$) activation thresholds. θ^* is chosen by Youden’s J on records 500–999 and applied to records 0–499. Exact top-1 recall is threshold-independent.

Query Formulation	Exact Top-1 Recall	Activated ($\theta = 0.60$)	Activated ($\theta^* = 0.55$)
Rewrite prompts (500)	99.2%	99.2%	99.2%
Paraphrase prompts (1,000)	91.0%	38.2%	53.2%
Neighborhood prompts (1,000), false activation	–	9.6%	16.1%
Neighborhood prompts (1,000), wrong-edit retrieval	–	2.4%	3.3%
ROC AUC (calibration split / evaluation split)		0.817 / 0.829	
HNSW candidate recall (10 candidates)		99.96%	

ROME protocol in four ways: one canonical prompt per edit, no multi-template key averaging or KL regularizer, a CounterFact-derived covariance corpus, and first-subtoken rather than full-sequence target scores. We therefore do not call it ROME and do not compare it to published ROME numbers.

On CounterFact-500, R1C attains an 85.2% efficacy score (35.6% argmax), 59.5% paraphrase score, and 70.1% neighborhood score, with control-text perplexity ratio 1.001 mean and 1.291 max. Its paraphrase score is compared with GRACE in the next subsection; its neighborhood score (70.1%) is a different metric from the read’s neighborhood false activation (16.1%) and the two are not directly comparable.

6.6. GRACE at Matched Scale

GRACE [15] is the memory-based editor closest in mechanism to the gated read: it stores a codebook of (key, value, ϵ) triples at one MLP layer, fires when the last-token input activation lies within ϵ of a stored key, and replaces that token’s layer output by the stored value. Its ϵ -ball plays the role of our activation threshold θ , with keys in a hidden-activation space rather than a sentence-embedding space. We re-implement GRACE in our own code following the algorithm and default hyper-parameters of the reference implementation, rather than running the authors’ released code, on GPT-2 (124M) at layer 9 of 12 (the relative depth of the layer used for GPT-2 XL), with $\epsilon_{\text{init}} \in \{1, 3, 10\}$, SGD value training (learn-

Table 5: CounterFact-500 on GPT-2 (124M) under one scorer. R1C and GRACE are parametric and memory-based edits; the read is the frozen-MiniLM external memory of Table 4, for which only activation rates are defined. Efficacy, paraphrase, and neighborhood scores compare $P(\text{new})$ with $P(\text{true})$ on the first subtoken. Base GPT-2 without any edit gives paraphrase score 28.1% and neighborhood score $\approx 71\%$.

Metric	R1C (reset)	GRACE reset ($\epsilon = 1$)	GRACE seq. ($\epsilon = 1$)	Read ($\theta^* = 0.55$)
Efficacy score / argmax	85.2 / 35.6	100.0 / 100.0	99.2 / 98.6	–
Rewrite activation	–	100.0	98.6	99.2
Paraphrase score / argmax	59.5 / –	28.1 / 0.0	28.1 / 0.0	–
Paraphrase activation	–	0.0	0.2	53.2
Neighborhood score	70.1	70.8	71.2	–
Neighborhood false activation	–	1.0	6.2	16.1
Control ppl. ratio (mean / max)	1.001 / 1.291	1.000 / 1.000	1.000 / 1.000	1.000 / 1.000
Stored state after 500 edits	1 matrix	–	481 entries	500 pairs

ing rate 1.0, 100 iterations) on the first-subtoken cross-entropy, and GRACE’s expand/split rule on key conflicts. Edits are applied to the same 500 records with the same scorer as R1C, under GRACE’s native sequential protocol (all 500 edits accumulated, evaluation after the last) and under per-record reset. Because the scorer compares first-subtoken probabilities rather than the full multi-token log-probabilities of the published CounterFact protocol, the numbers in Table 5 are comparable to each other and to Table 4, but not directly to efficacy, paraphrase, or neighborhood figures reported for ROME or GRACE elsewhere.

Table 5 gives the result. GRACE edits the rewrite prompt almost perfectly (100% efficacy with reset, 99.2% after 500 sequential edits) and leaves control perplexity exactly unchanged; the ratio of 1.000 is measured, but it is also forced by the mechanism, since no control-text activation falls inside any stored ϵ -ball and the wrapped layer is then the identity. It does not generalize at all: paraphrase activation is 0.0–0.2% and its paraphrase score, 28.1%, equals the unedited model’s. Widening ϵ_{init} to 3 or 10 does not help, because GRACE’s split rule shrinks conflicting balls to half the inter-key distance; after 500 edits the mean ϵ is 0.85, 1.01, and 1.02 for $\epsilon_{\text{init}} = 1, 3, 10$, paraphrase activation stays below 6%, and neighborhood false activation rises to 13–14%. In hidden-activation space the CounterFact paraphrases are simply not within reach of the rewrite key. The read operator, by contrast, activates on 53.2% of paraphrases at a 16.1% false-activation cost, and R1C’s weight update reaches 59.5% paraphrase score at the cost of a 35.6% argmax rate and a 1.29 worst-case perplexity ratio.

369 The comparison is informative about mechanism rather than about a winner.
370 GRACE and the read are both threshold-gated lookups, and they sit at opposite
371 ends of a locality trade-off that is set by the key space: hidden activations at one
372 token give near-zero false activation and near-zero generalization, sentence em-
373 beddings give moderate generalization at moderate false activation, and the ROC-
374 calibrated θ^* lets that trade-off be chosen from held-out data rather than fixed by
375 a single ϵ_{init} that the split rule then overrides. The comparison also shows what
376 the read gives up: GRACE’s value is a learned hidden-state replacement that pro-
377 duces the target argmax on 98.6–100% of rewrites, whereas the read’s value is a
378 token embedding that must be mixed into the final state (Section 6.9), and neither
379 mechanism edits the paraphrases that its gate does not reach.

380 6.7. *Dynamic Tri-Space Routing and Path Ablation*

381 We evaluate the tri-space architecture across three information paths: in-
382 context reasoning (Ω_{ctx} , 15 ephemeral premises supplied at query time), cor-
383 pus world knowledge (Ω_{corpus} , 50 stored facts), and parameter-resident ex-
384 pert outputs (Ω_{params} , 15 tasks served by a learned expert bank). The router
385 $\lambda(x) = \text{softmax}(\text{MLP}_{384 \rightarrow 128 \rightarrow 3}(x))$ acts on the raw MiniLM query embedding and
386 is trained jointly with the residual adapter, output embeddings, and expert bank
387 by AdamW (10^{-3} , weight decay 10^{-4} , up to 1200 epochs with patience 120) on
388 the sum of cross-entropy losses over the three task families. No routing labels are
389 used; route accuracy below is an emergent property of task supervision. Training
390 queries use family-specific wrappers; evaluation uses unseen strings under the
391 shared wrapper “Determine the output for:” across all three families, so that
392 surface form cannot identify the intended space.

393 Table 6 and Figure 1 show route accuracies of 100.0%, 82.4%, and 100.0%,
394 and task accuracies of 100.0%, 96.8%, and 100.0%. Ablating the intended space
395 reduces task accuracy to 0% across all five seeds. The corpus family is the weak
396 point: without route keywords the gate sends 30.5% of mass to the parameter path,
397 and on 17.6% of corpus queries the parameter gate is the largest. Task accuracy
398 nevertheless exceeds route accuracy (96.8% vs. 82.4%) because the parameter-
399 expert read, when wrongly favoured, returns a mixture of expert values that are
400 unrelated to the corpus target vocabulary; the smaller corpus-gated component
401 remains the only one aligned with the correct token, so the argmax survives. This
402 is a benign failure only because the two output vocabularies are disjoint here, and
403 would not hold if experts and corpus competed for the same tokens.

Table 6: Five-seed surface-matched routing and ablation (mean $\pm$ sample standard deviation). Route accuracy is the fraction whose largest gate selects the intended space.

Task Domain	Task Acc.	Route Acc.	Mean Gating Distribution $\lambda(x)$			Ablated Acc.
			λ_{ctx}	λ_{corpus}	λ_{params}	
In-Context Reasoning (Ω_{ctx})	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	68.0 $\pm$ 2.1	6.2 $\pm$ 0.9	25.8 $\pm$ 1.3	0.0 $\pm$ 0.0
Corpus World Knowledge (Ω_{corpus})	96.8 $\pm$ 1.8	82.4 $\pm$ 0.9	12.9 $\pm$ 0.7	56.6 $\pm$ 2.2	30.5 $\pm$ 1.7	0.0 $\pm$ 0.0
Parameter Expert Outputs (Ω_{params})	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	3.1 $\pm$ 0.2	3.1 $\pm$ 0.4	93.7 $\pm$ 0.5	0.0 $\pm$ 0.0

6.8. Anti-Atom Cancellation and Competitive Updating

Exact cancellation. Inserting an anti-atom ω^- with mass 1.0 under v^- at the key of a stored record realizes Proposition 1. On five sensitive-record queries in a 49-token output vocabulary, target probability in the external read falls from 99.998% to 2.04% = 1/49 (five-seed mean $2.0407 \pm 0.00003\%$), i.e. to the uniform distribution, and specificity on 35 unedited control facts stays at 100.0% (Table 7, top; Figure 2, right). This is a numerical confirmation of an algebraic identity and is reported for completeness; the informative experiment is the stress test below.

Approximate cancellation stress test. We perturb anti-atom keys, values, masses, or queries over 50 trials at each of background sizes 10, 100, and 1000 (3,000 trials) and compare the residual pair contribution with the bound of Proposition 2. The bound holds in all trials. At $N = 100$, key perturbations of ℓ_2 magnitude 0.01, 0.05, and 0.10 leave mean pair residuals of 4.5%, 72.1%, and 98.0% of the uncanceled contribution; because keys are unit-norm and the temperature is sharp, a perturbation of 0.05 already moves the anti-atom’s kernel weight far from the record’s. Query perturbations leave zero residual because identical keys receive identical kernel weights for every query. In practice, therefore, an anti-atom must be constructed from the stored key itself rather than re-encoded from text.

Competitive updating. Appending a contradicting atom with recency weight $w_{\text{new}} = 2.5$ overrides the old fact on only 44.0 $\pm$ 8.9% of records across five seeds (Table 7, bottom). Where it fails, the query is lexically closer to the original record than to the new one, and the recency weight does not compensate. Append-only

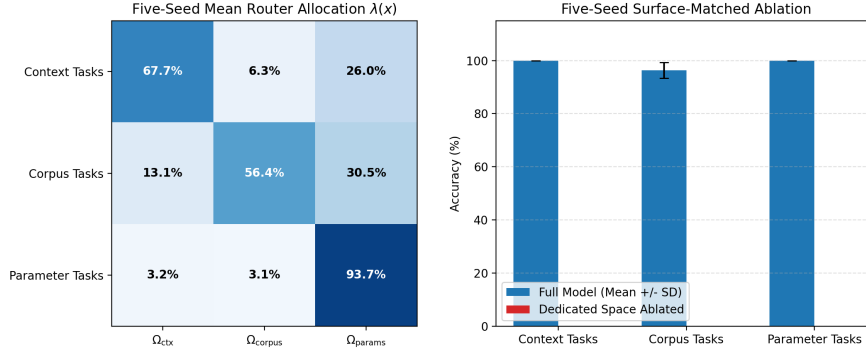


Figure 1: Five-seed surface-matched tri-space evaluation. Left: mean router allocation $\lambda(x)$. Right: mean task accuracy; intended-path ablation reduces accuracy to zero.

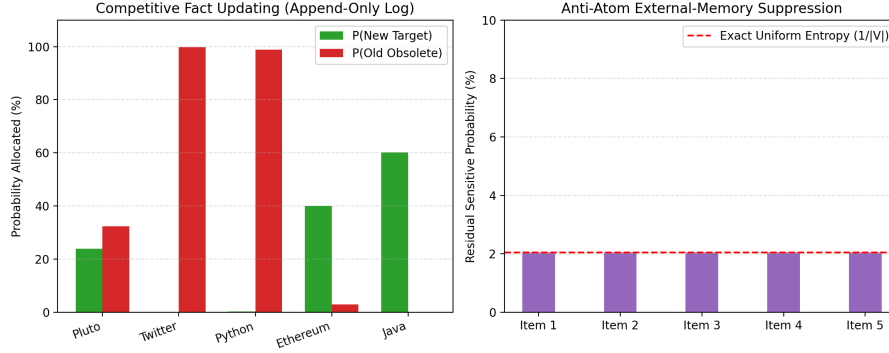


Figure 2: Signed atom spaces. Left: competitive updating in an append-only log ($w_{\text{new}} = 2.5$). Right: anti-atom cancellation drives residual target probability to uniform ($P = 2.04\% = 1/|\mathcal{V}|$).

426 updating without cancellation of the superseded record is therefore unreliable in
 427 this setting.

428 6.9. First-Token Steering of GPT-2 as a Retrieval Diagnostic

429 We couple Ω_{corpus} into the final hidden state of pretrained GPT-2 (124M [16])
 430 with cosine activation at $\theta \geq 0.60$:

$$\tilde{h}_t = \begin{cases} (1 - \lambda_{\text{read}})h_t + \lambda_{\text{read}}W_{\text{proj}}z_t, & \text{relevant and unconsumed,} \\ h_t, & \text{otherwise.} \end{cases}$$

431 Each memory value is the unit-normalized output embedding of the target’s first
 432 token. The construction is therefore circular by design: conditional on retrieval,

Table 7: Signed reads on a 49-token output vocabulary. Top: exact anti-atom cancellation (five records, identical outcome). Bottom: counterfactual updating by appending a contradicting atom with recency weight $w_{\text{new}} = 2.5$; three of five updates fail. Specificity on 35 clean background facts is 100.0% in all rows.

Target / Query	Pre-Edit P(Target)	Post-Edit P(Target)	Outcome
<i>Exact anti-atom cancellation (unit mass under v^-)</i>			
Five sensitive records (coordinates, diagnosis, patent, location, key)	99.998%	2.04% ($= 1/ \mathcal{V} $)	Cancelled to uniform
<i>Competitive updating ($w_{\text{new}} = 2.5$, append-only)</i>			
Ethereum $\rightarrow$ Proof-of-Stake	3.0% (old)	40.0% (new)	Success
Java $\rightarrow$ Oracle	0.01% (old)	60.1% (new)	Success
Pluto $\rightarrow$ Dwarf Planet	32.3% (old)	23.9% (new)	Failure: old fact retains argmax
Twitter $\rightarrow$ X Corp	99.8% (old)	0.04% (new)	Failure: lexical match to old record
Python $\rightarrow$ PyPy JIT	98.9% (old)	0.24% (new)	Failure: lexical match to old record

433 injecting the target embedding into the final hidden state is expected to raise the
434 target’s first-token probability. The experiment does not test whether the model
435 has “learned” a fact; it tests (a) whether the retrieval gate fires on the right prompts
436 and stays silent on controls, and (b) how large a hidden-state perturbation is needed
437 to flip the argmax. The probability ratios in Table 8 should be read in that light
438 and not as evidence of editing quality.

439 Table 8 and Figure 3 show first-token argmax correction on 10/10 canonical
440 prompts and 9/10 paraphrases; the single paraphrase failure is a gate miss (simi-
441 larity below θ), not a steering failure. On ten unrelated control texts the gate never
442 fires, so perplexity ratio is exactly 1.000.

443 The mechanism is limited to one-shot, first-token steering at the query bound-
444 ary. After the corrected first token is appended, the query representation at sub-
445 sequent steps drifts away from the record’s key, similarity falls below θ , and the
446 model reverts to its base distribution. If activation is instead held on across steps,
447 the persistent additive injection produces degenerate repetitions (e.g. “lead, tho-
448 rium, and lead-based materials”). Coherent multi-token generation would require
449 a sequence-level memory policy, which we do not provide.

450 Table 9 compares the read against naive fine-tuning (FT) and single-layer
451 ROME-lite on the same ten facts. Naive FT reaches 100% canonical efficacy

Table 8: Selected rows from the ten-fact greedy GPT-2 diagnostic. Because the injected value is the target’s own token embedding, large first-token probability gains are expected conditional on retrieval; the rows document gate behaviour and perturbation magnitude, not factual learning.

Factual Target Task	Target Probability $P(y^*)$		Ratio	Top Predicted Token	
	Base GPT-2	Atom-Space		Base GPT-2	Atom-Space
QUIC Transport Protocol (UDP)	2.64%	70.00%	26.5×	the	UDP
Milky Way Black Hole (Sagittarius)	0.47%	64.33%	136.3×	after	Sag
Safe Systems Language (Rust)	0.03%	32.99%	1116.0×	a	Rust
Uranium Radiative Decay (lead)	0.87%	9.80%	11.3×	uranium	lead
CRISPR Bacterial Defense (viruses)	2.25%	38.41%	17.1×	infect.	viruses

452 but changes predictions on 41.1% of neighbouring facts; ROME-lite preserves
453 locality (87.8%) but generalizes less to paraphrases (40% vs. 90%). The read
454 leaves weights untouched, so its specificity failure mode is false gate activation
455 rather than parameter interference; that rate was not measured on the ten facts and
456 is 16.1% at θ^* on CounterFact-500. The three columns therefore trade off different
457 failure modes and the table should not be read as a ranking.

458 *CounterFact steering at $N = 50$.* We repeat the diagnostic on 50 CounterFact
459 records at the calibrated threshold $\theta^* = 0.55$, with 2,000 bootstrap resamples
460 for intervals. Base GPT-2 assigns $P(\text{new}) > P(\text{true})$ on 16.0% [6.0%, 26.0%]
461 of rewrite prompts and 22.0% [12.0%, 34.0%] of paraphrases, with 0.0% target
462 argmax accuracy. The gate fires on 100.0% of rewrites and 58.0% of paraphrases,
463 consistent with the activation rates of Table 4. Conditional on activation the steered
464 model attains 100.0% rewrite efficacy and argmax accuracy, and overall paraphrase
465 score 66.0% [52.0%, 78.0%] with 56.0% [42.0%, 70.0%] exact argmax. Because
466 paraphrase steering is bounded above by paraphrase activation, the gap between
467 58% activation and 91% exact top-1 recall (Table 4) is where the useful headroom
468 lies; the injection itself is not the bottleneck. This is a retrieval diagnostic and not
469 an end-to-end comparison with RAG or with model editors.

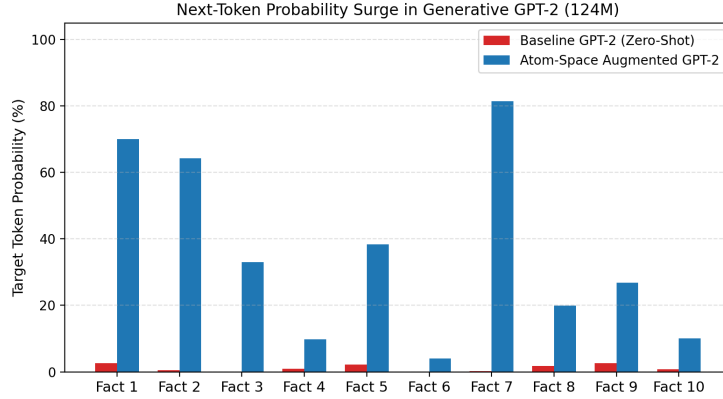


Figure 3: First-token target probabilities in GPT-2 (124M) before and after target-embedding intervention.

7. Discussion and Related Work

The computational elements unified under our framework have extensive histories across distinct areas of machine learning. External corpus reads $R(x; \Omega_{\text{corpus}})$ correspond to nearest-neighbor language models and retrieval-augmented architectures, such as k NN-LM [7, 17], RETRO [8], and memory network designs [18, 12]. Parameter-expert reads $R(x; \Omega_{\text{params}})$ encapsulate modular computation, including sparsely gated Mixture-of-Experts [9, 19, 11] and dynamic adapter routing banks [10]. Finally, standard self-attention [1] constitutes an in-context read $R(x; \Omega_{\text{ctx}})$ over sequence coordinates, whose continuous formulation relates naturally to kernel attention smoothing and linear attention mechanisms [13, 20].

7.1. Relation to Memory-Based and Non-Parametric Model Editing

The atom-space read operator connects directly to non-parametric, semi-parametric, and memory-augmented model editing strategies.

Semi-Parametric Edit Retrieval and Correction (SERAC) [21] caches edits externally and employs an auxiliary classifier to route queries between a base language model and a separate counterfactual sequence-to-sequence model. The read operator shares the objective of keeping base weights frozen, but its retrieval is a convex combination $R(x; \Omega)$ that combines linearly within a shared representation space, avoiding a second autoregressive model. The trade-off is direct: SERAC retains full sequence-level generation by delegating altered queries to the counterfactual network, while the read performs a single-pass vector addition at the

Table 9: Naive full fine-tuning and single-layer rank-one editing (ROME-lite) versus the relevance-gated read on the same ten GPT-2 facts, paraphrases, and controls. The read’s cross-fact specificity was not measured on these ten facts; its analogue at scale is the neighborhood false-activation rate of Table 4.

Metric	Atom-Space Read	Naive Fine-Tuning (FT)	ROME-Lite
Canonical efficacy (first-token argmax)	100% (10/10)	100% (10/10)	100% (10/10)
Paraphrase generality (first-token argmax)	90% (9/10)	70% (7/10)	40% (4/10)
Neighborhood specificity (other 9 facts unchanged)	not measured (weights unchanged; gate false activation 9.6–16.1% on CounterFact-500)	58.9% (mean)	87.8% (mean)
Control-text perplexity ratio	1.000 (exact)	0.982 mean, 1.022 max	0.996 mean, 1.004 max
Mechanism	Relevance-gated read	Full-parameter AdamW	Rank-one weight update

491 representation layer and, as Section 6.9 shows, does not generate coherent multi-
 492 token continuations.

493 GRACE [15] inserts norm-constrained key-value codebook patches into inter-
 494 mediate transformer activations, with ϵ -ball activation around stored keys. The
 495 activation gate θ of Section 6.4 plays the same role as GRACE’s ϵ ; the differences
 496 are that our keys live in an external sentence-embedding space rather than a hid-
 497 den layer, that cancellation is expressed on the reference measure rather than by
 498 codebook deletion, and that we report ROC-calibrated rather than fixed thresholds.
 499 Section 6.6 runs GRACE on the same 500 records with the same GPT-2 check-
 500 point and scorer and finds that the two gated lookups occupy opposite ends of a
 501 generalization–locality trade-off set by the key space. We have not run matched
 502 comparisons with SERAC, MEMIT [5], MEND [22], WISE [23], or AlphaEdit
 503 [24], and nothing here should be read as a claim of superiority over them.

504 Interpolation between context attention and external memory stores has also
 505 been investigated in Memorizing Transformers [25] and adaptive k NN-LM [17].
 506 The atom-space view places token context, dense external corpora, and parameter-
 507 expert banks under one read abstraction with separate normalization, which is the
 508 calibration principle of Section 3.2.

509 Finally, signed atom spaces interface with machine unlearning. SISA [26] and
 510 Fisher-based certified removal adjust weights to eliminate the statistical influence

511 of target data. Signed atom spaces cancel only in the read output, leave pretrained
512 weights unmodified, and retain both the record and its anti-atom in storage; they
513 are a mechanism for external-memory retraction, not for parametric unlearning.

514 7.2. *Limitations*

515 The atom-space view is a shared notation; every result here is a finite-sum
516 statement and none exploits a non-atomic measure. It does not imply that attention,
517 RAG, and MoE are interchangeable. On the theoretical side, the Lipschitz-margin
518 condition is vacuous on the benchmark studied and its finite-sample L estimates
519 do not establish the global premise, while the truncation certificate requires all
520 N scores and selects 52–86% of a synthetic corpus at $N = 25,000$; both are re-
521 ported as calibrations of how loose worst-case arguments are, not as operational
522 tools. The 200-fact benchmark and the tri-space routing tasks are templated; the
523 insertion metrics are saturated, frozen 1-NN matches the trained read, and rout-
524 ing failures are benign only because the corpus and expert output vocabularies
525 are disjoint. In the GPT-2 steering study the injected value is the target’s own
526 token embedding, so first-token gains are expected conditional on retrieval; the
527 experiment measures gate behaviour rather than factual learning, and multi-token
528 generation drifts or degenerates. The R1C reference is not a ROME reproduc-
529 tion, and the GRACE comparison is our own re-implementation on GPT-2 small
530 at a proportionally chosen layer with a first-subtoken value objective rather than
531 the authors’ code in its published GPT-2 XL hallucination setup; all CounterFact
532 scores in this paper use first-subtoken probability comparisons and are therefore
533 internally consistent but not numerically interchangeable with published ROME
534 or GRACE results. No comparison with SERAC, MEMIT, MEND, WISE, or Al-
535 phaEdit is provided, and the retrieval read is scored on activation and first-token
536 metrics rather than as an end-to-end editor. All experiments use GPT-2 (124M);
537 the argument does not depend on scale, but the effect sizes have not been checked
538 on a larger model. Finally, anti-atoms retain both records, cancel only the exter-
539 nal read, require the exact stored key, and do not remove knowledge from dense
540 parameters.

541 8. **Conclusion**

542 We have used a single read operator over valued atom spaces as a common
543 notation for attention, dense retrieval, and expert routing, and used it to pose three
544 questions on a shared footing: when an inserted record is retrieved, when a read
545 can be truncated, and when a record can be cancelled. The answers are mixed

and, we argue, informative. A natural Lipschitz-margin insertion condition certifies nothing on a corpus where insertion succeeds on every record. An exact softmax tail certificate holds by construction but licenses far less truncation than practitioners use. Matched anti-atoms cancel exactly, but a key perturbation of 0.05 leaves 72% of the contribution. On CounterFact-500, held-out ROC calibration of the activation threshold buys 15 points of paraphrase activation for 6.5 points of false activation, and the remaining gap to 91% exact top-1 recall is where retrieval-based editing has headroom. Run on the same records with the same scorer, GRACE’s hidden-activation codebook reaches 99–100% rewrite efficacy with no measurable side effects but fires on almost no paraphrases, which places the two gated lookups at opposite ends of a trade-off that is governed by the key space rather than by the gating rule. The next steps are local or directional insertion guarantees that do not rely on a global Lipschitz premise, tail bounds under explicit score-distribution assumptions, and matched comparisons with MEMIT and SERAC under their published protocols.

CRedit Authorship Contribution Statement

Farshid Mossaiby: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The complete codebase, benchmark definitions, experimental suites, and reproduction scripts are publicly available at <https://github.com/mossaiby/RoAS>.

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author used Google Gemini to assist with drafting prose, checking the algebra in the proofs of Section 5, designing

the experimental pipeline, and manuscript formatting. All theorem statements, proofs, code, and experimental results were verified by the author, who takes full responsibility for the content of the publication.

References

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, in: Advances in Neural Information Processing Systems, Vol. 30, 2017, pp. 5998–6008.
- [2] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, D. Amodei, Language models are few-shot learners, in: Advances in Neural Information Processing Systems, Vol. 33, 2020, pp. 1877–1901.
- [3] M. Geva, R. Schuster, J. Berant, O. Levy, Transformer feed-forward layers are key-value memories, in: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, 2021, pp. 5484–5495.
- [4] K. Meng, D. Bau, A. Andonian, Y. Belinkov, Locating and editing factual associations in GPT, in: Advances in Neural Information Processing Systems, Vol. 35, 2022, pp. 17359–17372.
- [5] K. Meng, A. S. Sharma, A. Andonian, Y. Belinkov, D. Bau, [Mass-Editing Memory in a Transformer](#) (2023). [arXiv:2210.07229](#).
URL <https://arxiv.org/abs/2210.07229>
- [6] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, D. Kiela, Retrieval-augmented generation for knowledge-intensive NLP tasks, in: Advances in Neural Information Processing Systems, Vol. 33, 2020, pp. 9459–9474.
- [7] U. Khandelwal, O. Levy, D. Jurafsky, L. Zettlemoyer, M. Lewis, [Generalization through memorization: Nearest neighbor language models](#) (2020). [arXiv:1911.00172](#).
URL <https://arxiv.org/abs/1911.00172>

- [8] S. Borgeaud, A. Mensch, J. Hoffmann, T. Cai, E. Rutherford, K. Millican, G. B. V. D. Driessche, J.-B. Lespiau, B. Damoc, A. Clark, D. de Las Casas, A. Guy, J. Menick, R. Ring, T. Hennigan, S. Huang, L. Maggiore, C. Jones, A. Cassirer, A. Brock, M. Paganini, G. Irving, O. Vinyals, S. Osindero, K. Simonyan, J. W. Rae, E. Elsen, L. Sifre, Improving language models by retrieving from trillions of tokens, in: Proceedings of the 39th International Conference on Machine Learning, Vol. 162, PMLR, 2022, pp. 2206–2240.
- [9] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, J. Dean, [Outrageously large neural networks: The sparsely-gated mixture-of-experts layer](#) (2017). [arXiv:1701.06538](#).
URL <https://arxiv.org/abs/1701.06538>
- [10] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, [LoRA: Low-rank adaptation of large language models](#) (2021). [arXiv:2106.09685](#).
URL <https://arxiv.org/abs/2106.09685>
- [11] A. Q. Jiang, A. Sablayrolles, A. Roux, A. Mensch, B. Savary, C. Bamford, D. S. Chaplot, D. de las Casas, E. B. Hanna, F. Bressand, G. Lengyel, G. Bour, G. Lample, L. R. Lavaud, L. Saulnier, M.-A. Lachaux, P. Stock, S. Subramanian, S. Yang, S. Antoniak, T. L. Scao, T. Gervet, T. Lavril, T. Wang, T. Lacroix, W. E. Sayed, [Mixtral of experts](#) (2024). [arXiv:2401.04088](#).
URL <https://arxiv.org/abs/2401.04088>
- [12] G. Lample, A. Sablayrolles, M. Ranzato, L. Denoyer, H. Jégou, Large memory layers with product keys, in: Advances in Neural Information Processing Systems, Vol. 32, 2019, pp. 8546–8557.
- [13] Y.-H. H. Tsai, S. Bai, M. Yamada, L.-P. Morency, R. Salakhutdinov, Transformer dissection: An unified understanding for transformer’s attention via the lens of kernel, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing, 2019, pp. 4344–4353.
- [14] A. F. T. Martins, A. Farinhas, M. Treviso, V. Niculae, P. M. Q. Aguiar, M. A. T. Figueiredo, Sparse and continuous attention mechanisms, in: Advances in Neural Information Processing Systems, Vol. 33, 2020, pp. 20989–21001.

- [15] T. Hartvigsen, S. Sankaranarayanan, H. Palangi, Y. Kim, M. Ghassemi, Aging with GRACE: Lifelong model editing with discrete key-value adaptations, in: Advances in Neural Information Processing Systems, Vol. 36, 2023, pp. 56734–56747.
- [16] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever, Language models are unsupervised multitask learners, Tech. rep., OpenAI (2019).
- [17] J. He, G. Neubig, T. Berg-Kirkpatrick, Efficient nearest neighbor language models, in: Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, 2021, pp. 5703–5714.
- [18] S. Sukhbaatar, A. Szlam, J. Weston, R. Fergus, [End-to-end memory networks](#) (2015). [arXiv:1503.08895](#).
URL <https://arxiv.org/abs/1503.08895>
- [19] W. Fedus, B. Zoph, N. Shazeer, Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, Journal of Machine Learning Research 23 (120) (2022) 1–39.
- [20] A. Katharopoulos, A. Vyas, N. Pappas, F. Fleuret, Transformers are RNNs: Fast autoregressive transformers with linear attention, in: Proceedings of the 37th International Conference on Machine Learning, 2020, pp. 5156–5165.
- [21] E. Mitchell, C. Lin, A. Bosselut, C. D. Manning, C. Finn, Memory-based model editing at scale, in: Proceedings of the 39th International Conference on Machine Learning, 2022, pp. 15817–15831.
- [22] E. Mitchell, C. Lin, A. Bosselut, C. Finn, C. D. Manning, [Fast model editing at scale](#) (2022). [arXiv:2110.11309](#).
URL <https://arxiv.org/abs/2110.11309>
- [23] P. Wang, Z. Li, N. Zhang, Z. Xu, Y. Yao, Y. Jiang, P. Xie, F. Huang, H. Chen, [WISE: Rethinking the knowledge memory for lifelong model editing of large language models](#) (2024). [arXiv:2405.14768](#).
URL <https://arxiv.org/abs/2405.14768>
- [24] J. Fang, H. Jiang, K. Wang, Y. Ma, S. Jie, X. Wang, X. He, T. seng Chua, [Alphaedit: Null-space constrained knowledge editing for language models](#) (2025). [arXiv:2410.02355](#).
URL <https://arxiv.org/abs/2410.02355>

- 672 [25] Y. Wu, M. N. Rabe, D. Hutchins, C. Szegedy, [Memorizing transformers](#)
673 (2022). [arXiv:2203.08913](#).
674 URL <https://arxiv.org/abs/2203.08913>
- 675 [26] L. Bourtoutle, V. Chandrasekaran, C. A. Choquette-Choo, H. Jia, A. Travers,
676 B. Zhang, D. Lie, N. Papernot, Machine unlearning, in: 2021 IEEE Sympo-
677 sium on Security and Privacy (SP), 2021, pp. 141–159.